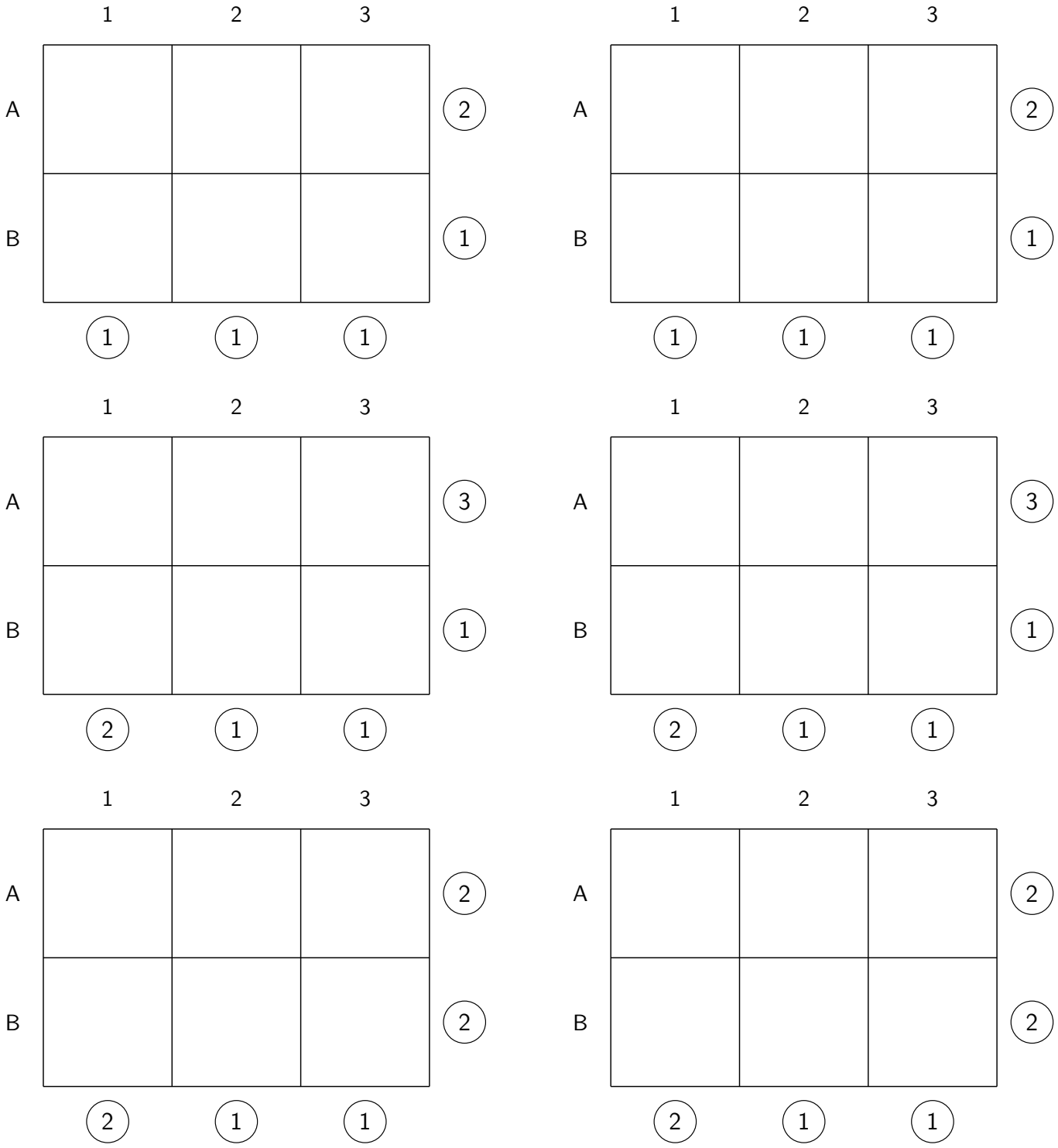
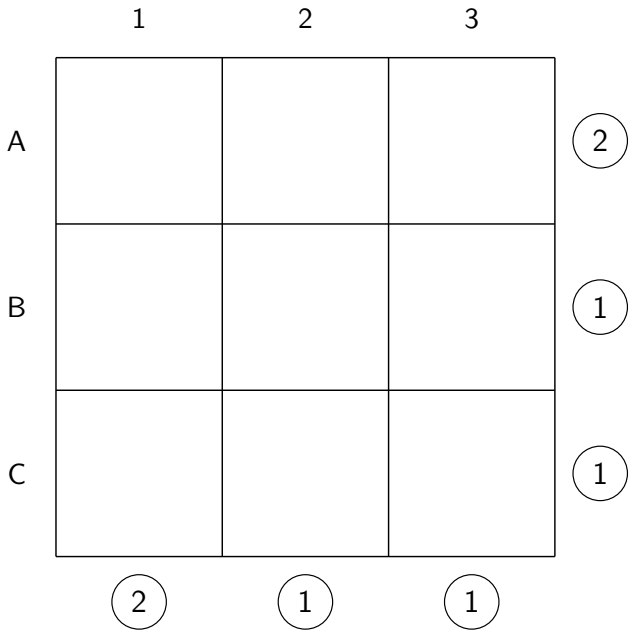
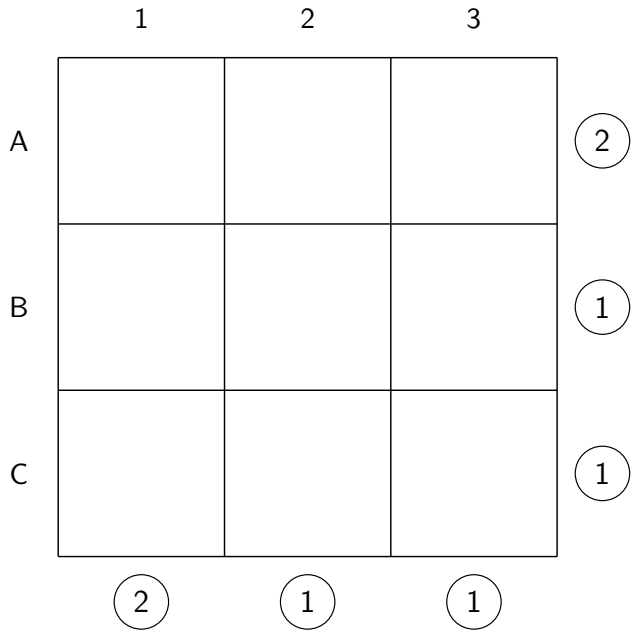
Each circled number counts the counters in its whole row or column. Put at most one counter in a cell. Two pictures are different if any cell is different.

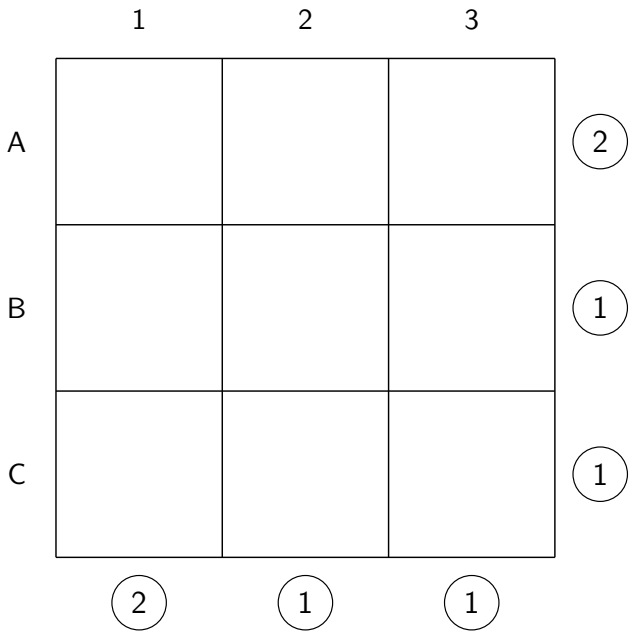
Problem 1: For each pair of grids, make two different pictures that match the counts, if you can.

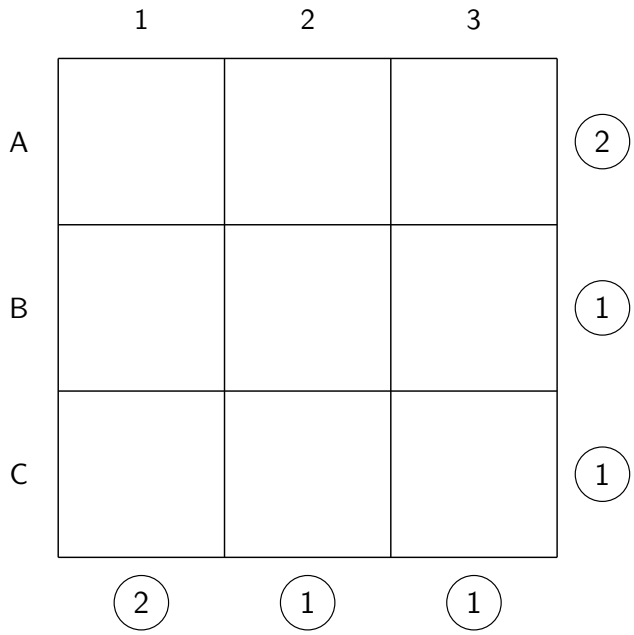


Problem 2: Find and draw every different picture with these counts. How can you be sure your list is complete?

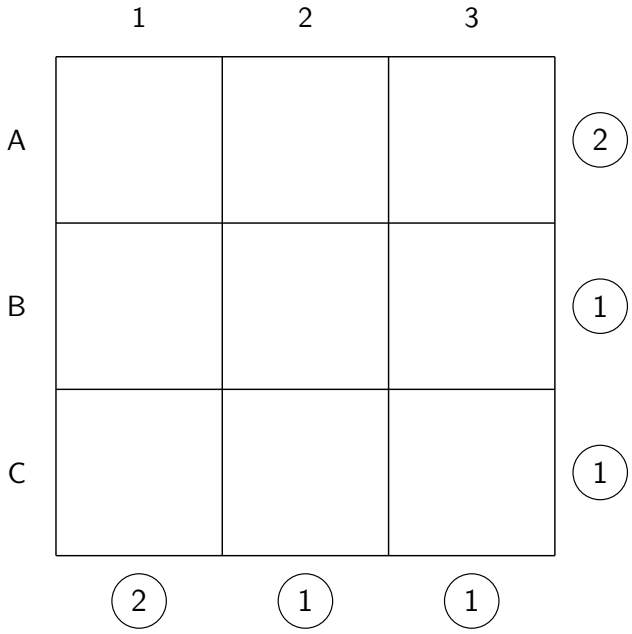


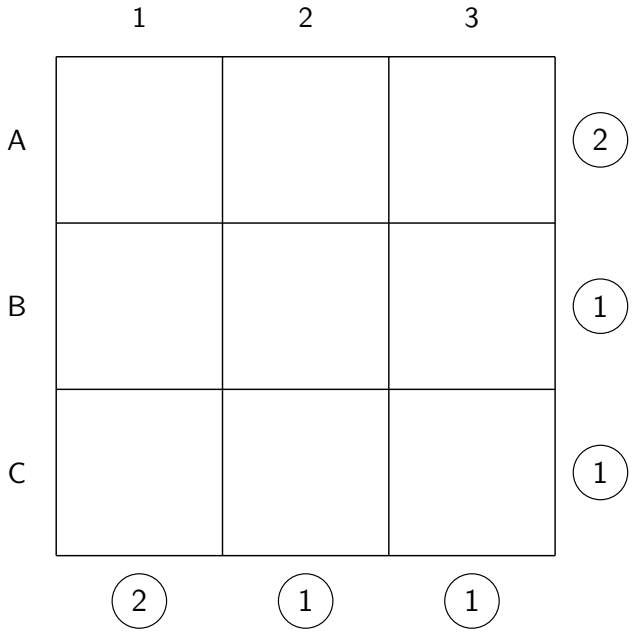




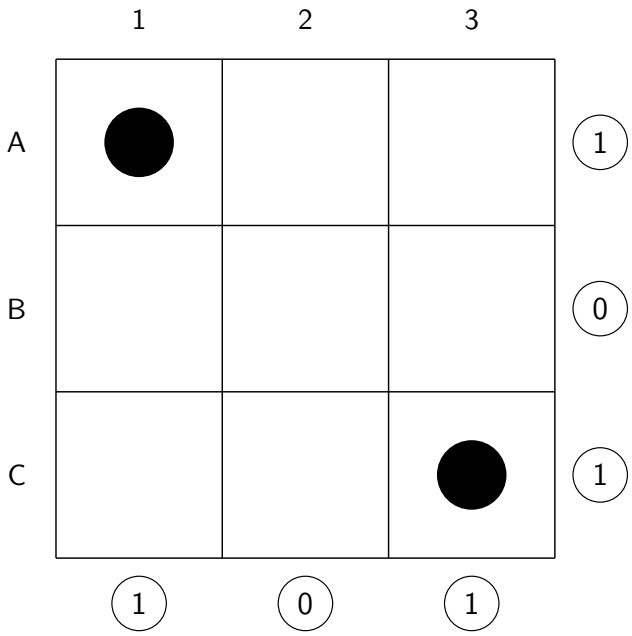


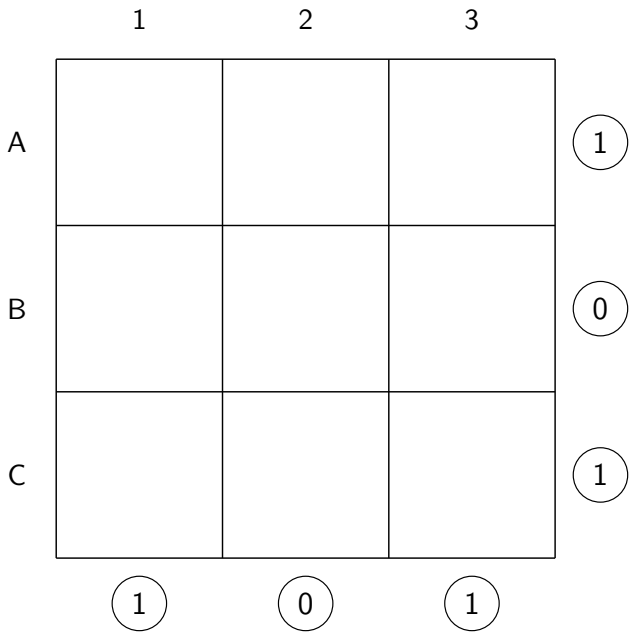
Problem 2 (continued)



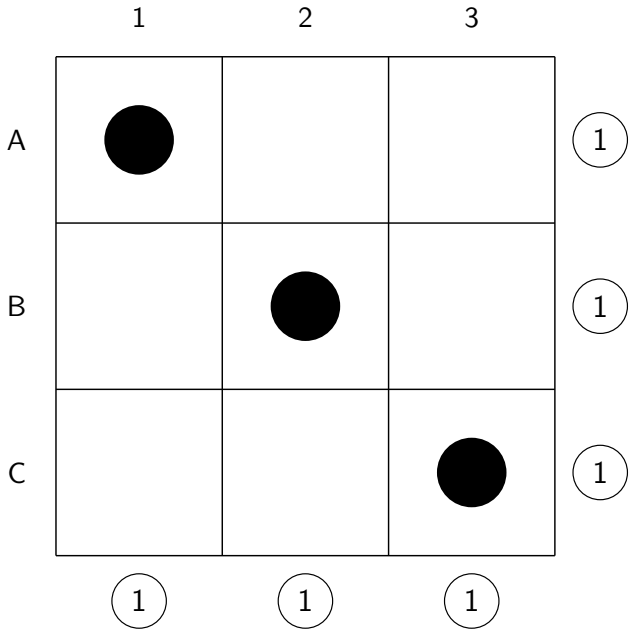


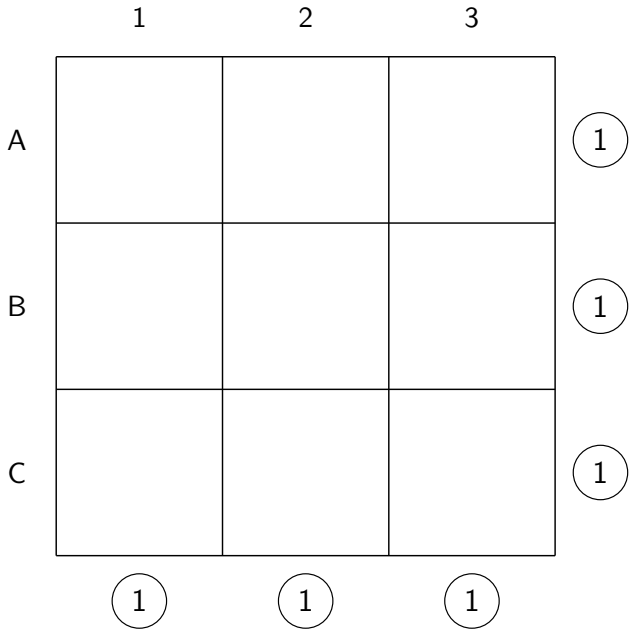
Problem 3: Move exactly two counters in each filled picture to make a different picture with the same counts. Find and draw every way.

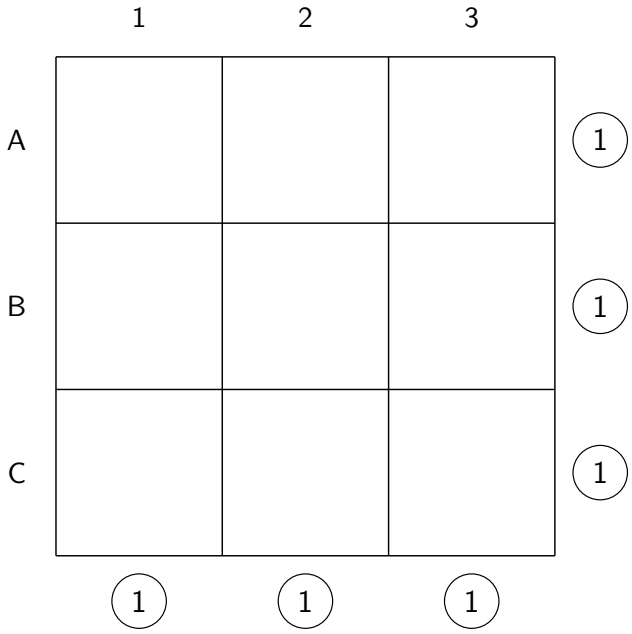


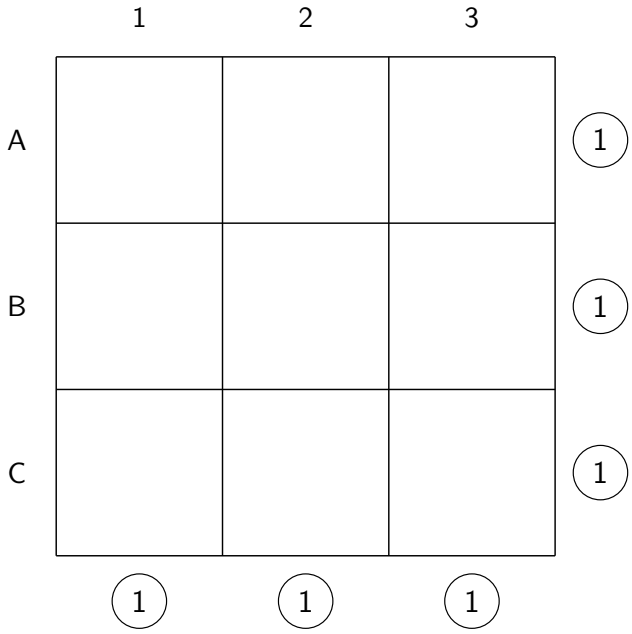


Problem 3 (continued)

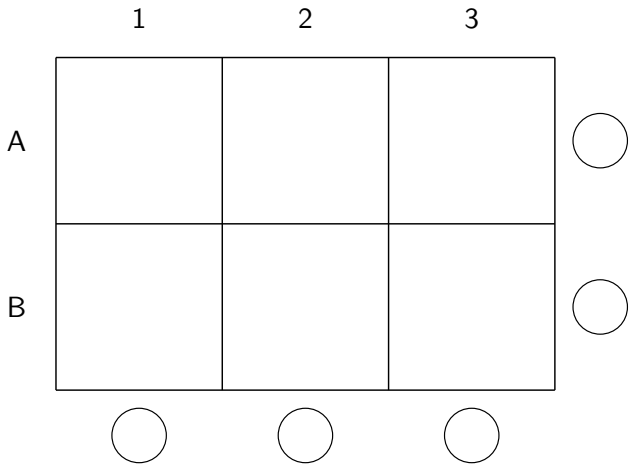


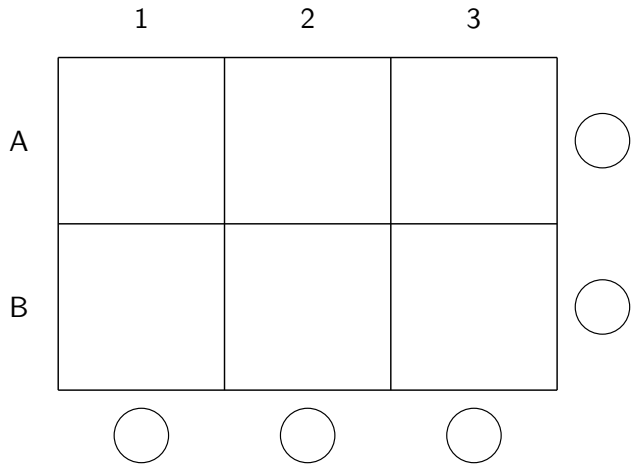


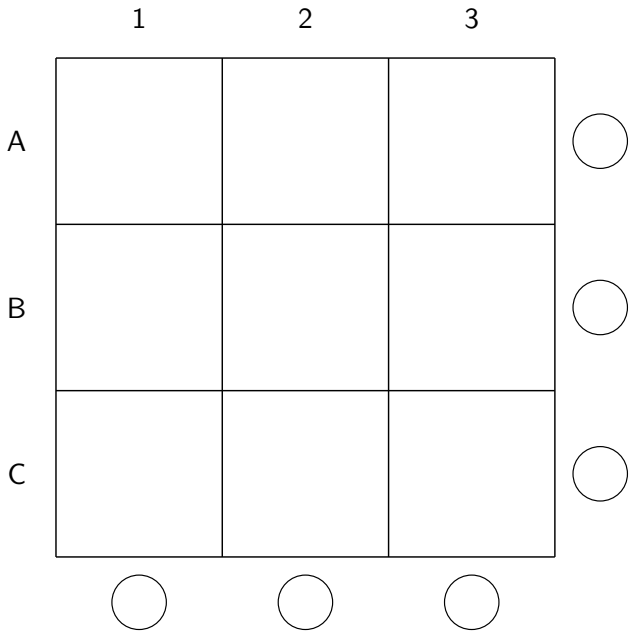


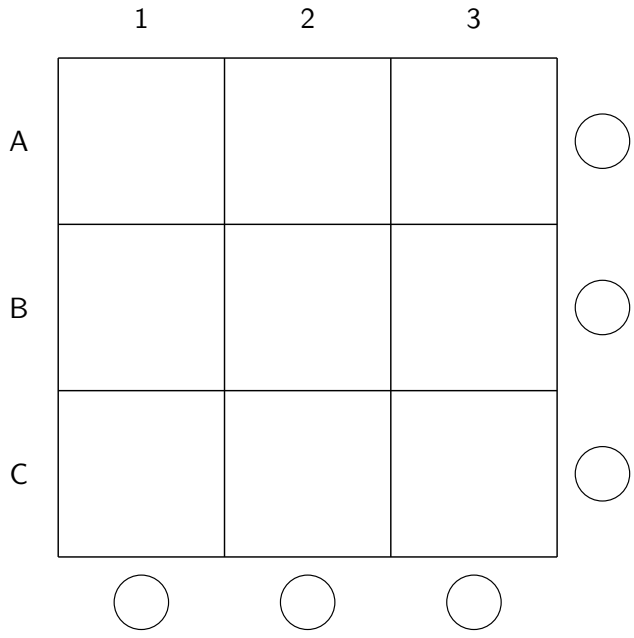


Problem 4: Use four counters on each left grid and write its counts. Ask a partner to make a different picture with the same counts on the right grid. Can you choose your picture so that this is impossible? Explain why.



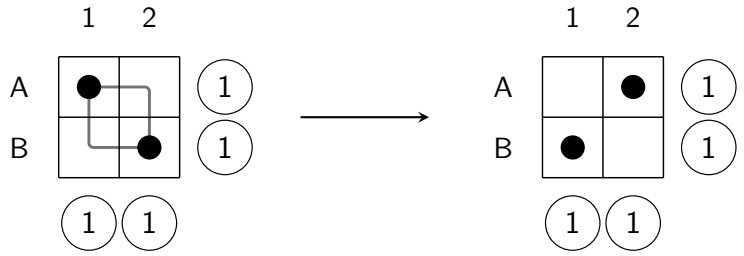


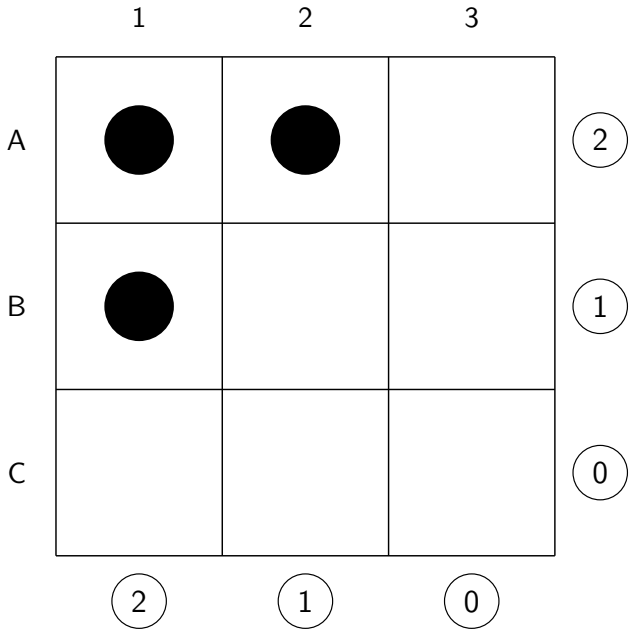


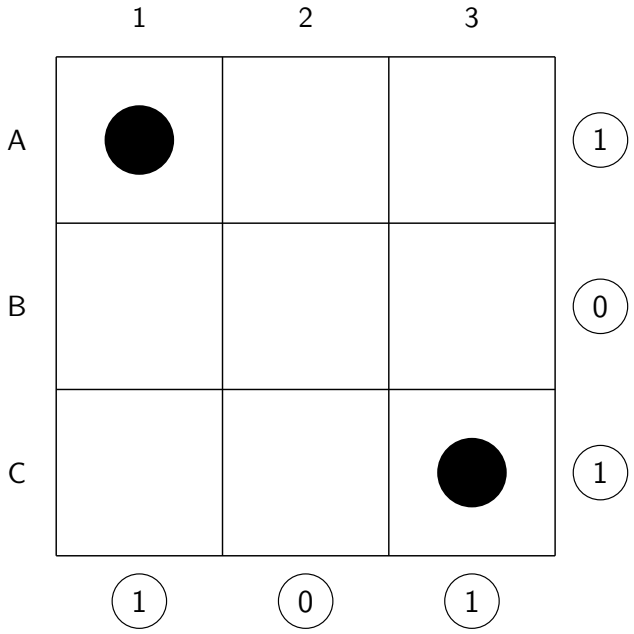


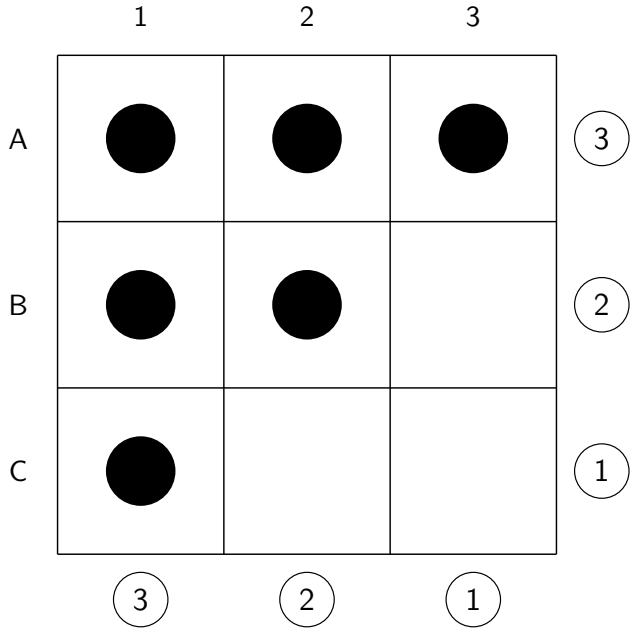
Problem 5: A switch moves the two counters to the empty corners shown; every other cell stays fixed. The corners may use any two rows and columns. Find every switch in each picture and draw its rectangle. Which pictures have none?

One switch:

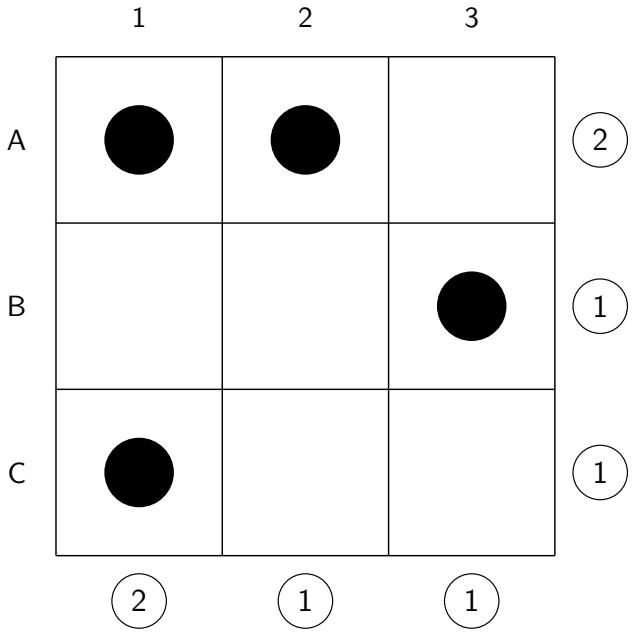


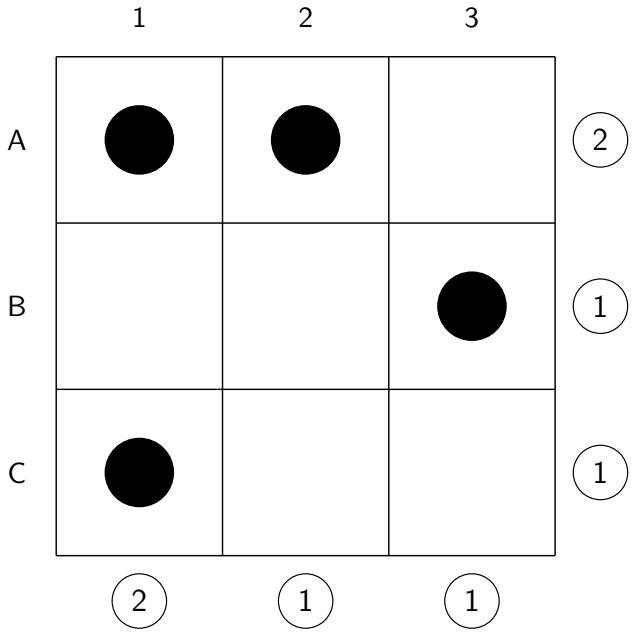


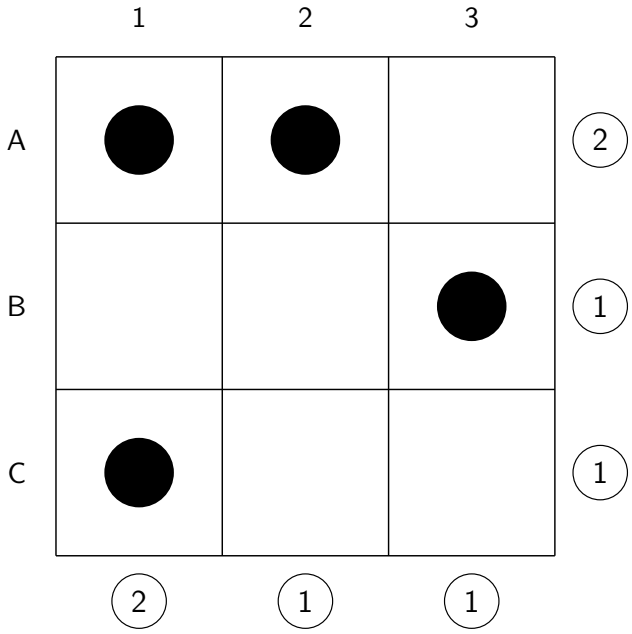


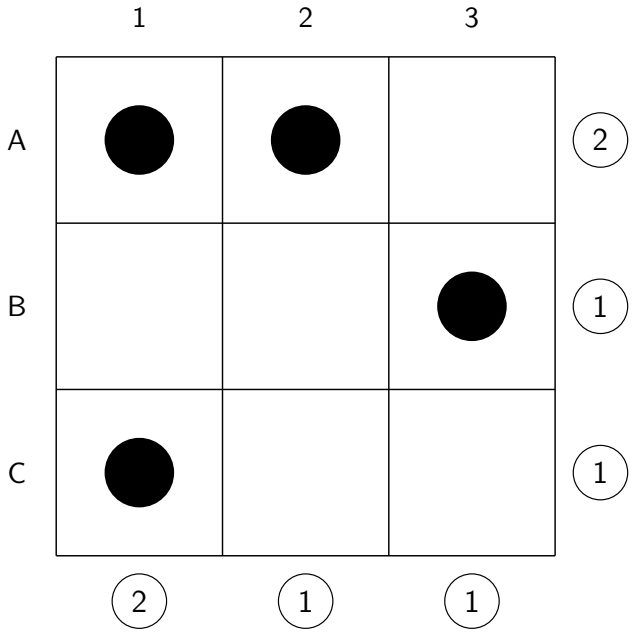


Problem 5 (continued)









Problem 6: Make a picture on the middle grid that takes exactly one switch to reach from the top picture. Make a picture on the bottom grid that takes exactly two switches and cannot be reached in one. Explain why one switch cannot reach it.

